Name:_____ Period:_____
Earth and Space Science

DENSITY, PHASE CHANGES, AND MATERIAL BEHAVIOR LEARNING GUIDE

Learning Intention
Today we will be learning how temperature and phase changes affect the density and behavior of Earth materials so we can explain why magma rises, why lava hardens into rock, why ice floats, and why Earth's layers are arranged the way they are.

Success Criteria
I can… ● Explain how heating and cooling change a material's density and why less dense material rises. ● Describe each phase change and whether energy is absorbed or released. ● Use density to explain the order of Earth's layers and the difference between oceanic and continental crust.

Hawai‘i Volcanoes: Rock That Flows

Hawai‘i Volcanoes National Park, on the Island of Hawai‘i, is home to Kīlauea, one of the most active volcanoes on Earth. Lava erupts at more than 1,100°C and flows downhill like a thick liquid. As it cools, it hardens into a dark rock called basalt. In other parts of the park, rainwater that soaks into the hot ground turns to steam and rises out of cracks called steam vents. In this guide, Hawai‘i is our example: here you can see matter change from liquid to solid and liquid to gas, and see how temperature and density decide where Earth materials go. As you read, keep this question in mind: *How do heat and density decide whether Earth materials rise, sink, melt, or harden?*

Density Review

Remember from Unit 1 that **density** is the amount of mass in a given volume of a material.

- **Density = mass ÷ volume.** Density is usually measured in grams per cubic centimeter (g/cm³). The reference tables do not list this formula, so you need to remember it.
- **Size does not matter.** Breaking a sample into pieces does not change its density. A small chip of basalt and a large boulder of the same basalt have the same density.
- **Less dense floats, more dense sinks.** A material that is less dense than the fluid around it will rise or float. A material that is more dense will sink.

Check for Understanding

1. A sample of Hawaiian basalt has a mass of 60 g and a volume of 20 cm³. Calculate its density.

_____ g/cm³

_____ 2. The basalt sample from question 1 is broken into two equal pieces. Compared to the whole sample, the density of each piece is
(1) half as much
(2) the same
(3) twice as much
(4) impossible to determine

Temperature and Density

When most materials are heated, their particles move faster and spread farther apart. The material **expands**: its mass stays the same, but its volume increases, so its density **decreases**. When a material cools, it **contracts** and its density **increases**.

This is why hot material rises and cool material sinks, the same idea behind the convection currents in the last learning guide. Hot **magma** (melted rock underground) is less dense than the solid rock around it, so it rises toward the surface. Scientists think a column of hot, rising mantle rock beneath Hawai‘i supplies the magma for its volcanoes.

Check for Understanding

_____ 1. As a sample of rock is heated, its volume and density change. Which row best describes these changes?
(1) volume increases and density increases
(2) volume increases and density decreases
(3) volume decreases and density increases
(4) volume decreases and density decreases

2. Explain why magma rises toward Earth's surface. Use the word *density*.

_____

_____

Phase Changes and Energy

Earth materials can be solid, liquid, or gas. A **phase change** happens when a material changes from one of these states to another. Every phase change involves energy. When energy is **absorbed**, particles move faster and break away from each other. When energy is **released**, particles slow down and lock together. When magma or lava freezes, its particles lock together to form mineral crystals. This is called **crystallization**.

Phase change	Change of state	Energy	Hawai‘i example
Melting	solid → liquid	absorbed	Rock deep underground melts to form magma
Freezing (crystallization)	liquid → solid	released	Lava cools and hardens into basalt
Evaporation	liquid → gas	absorbed	Rainwater on hot ground turns into steam
Condensation	gas → liquid	released	Steam from a vent cools into a cloud of tiny water droplets

Check for Understanding

1. Write **A** if energy is absorbed or **R** if energy is released.

_____ a. A lava flow hardens into basalt.

_____ b. Rock deep underground melts into magma.

_____ c. Steam from a vent turns into mist.

_____ d. Rain hitting hot lava boils away.

_____ 2. During which phase change do mineral crystals form in cooling lava?
(1) melting
(2) freezing
(3) evaporation
(4) condensation

Water Is Unusual

Most materials are more dense as solids than as liquids. Water is an exception. When water freezes, its molecules lock into a pattern that takes up more space, so water **expands** by about 9%. This makes ice **less**

dense than liquid water, which is why ice floats. Even in Hawai‘i, snow and ice form on the summit of Mauna Kea in winter.

Because water expands when it freezes, water that seeps into a crack in a rock and then freezes pushes the sides of the crack apart. Over many freezes, this can break the rock. You will study this process, called frost wedging, in a later unit.

Check for Understanding

_____ 1. Ice floats on liquid water because ice
(1) has more mass than liquid water
(2) is less dense than liquid water
(3) is warmer than liquid water
(4) absorbs more sunlight than liquid water

2. Rainwater seeps into a crack in a rock near the summit of Mauna Kea and freezes overnight. Predict what will happen to the crack and explain why.

_____

_____

Density and Earth's Layers

Rock in the mantle is solid, but it is not rigid like rock at the surface. Deep inside Earth, high temperature and pressure allow solid rock to flow very slowly, only a few centimeters per year. This slow flow is what makes mantle convection possible.

Early in Earth's history, much of the planet was hot enough to melt. The **densest** materials, such as iron and nickel, sank toward the center, and less dense materials rose toward the surface. This is why Earth has layers. From the outside in, they are the **crust**, **mantle**, **outer core**, and **inner core**. The crust is the least dense layer, and the inner core is the most dense. Scientists know about these layers from **seismic waves**, earthquake waves that change speed and direction as they pass through layers of different density. You will study this evidence in Unit 5.

There are two types of crust. **Oceanic crust** is made mostly of **basalt**, a dark rock with a density of about 3.0 g/cm³. The Hawaiian Islands are built from basalt. **Continental crust** is made mostly of **granite**, a lighter-colored rock with a density of about 2.7 g/cm³. Because oceanic crust is more dense, it sinks beneath continental crust when the two collide.

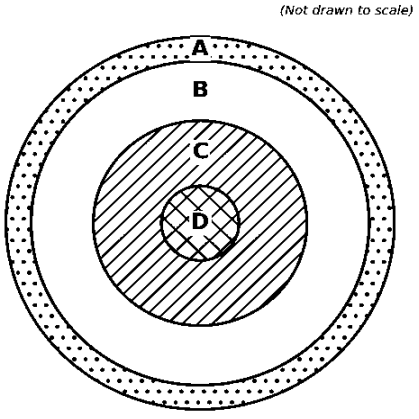
Check for Understanding: Model It

The diagram represents Earth's layers.

1. **Write the name** of each layer next to its letter. Then **circle the letter** of the most dense layer.

A: _____ B: _____

C: _____ D: _____



_____ 2. Compared to continental crust, oceanic crust is
(1) less dense and made mostly of granite
(2) less dense and made mostly of basalt
(3) more dense and made mostly of granite
(4) more dense and made mostly of basalt

3. Explain how the mantle can be made of solid rock but still flow.

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_____

ESSRT Connection

Use the **Generalized Model of Earth's Interior Structure** (page 11) in the 2026 Revised Edition Reference Tables for Earth and Space Sciences.

1. Which layer of Earth has the greatest density?

_____

_____ 2. According to the model, as depth increases from the crust to the center of Earth, density
(1) increases
(2) decreases
(3) remains the same
(4) decreases, then increases

3. Which layer is located directly between the mantle and the inner core?

_____

Cluster Set 1

Base your answers to questions 1 through 4 on the information and data table below and on your knowledge of Earth and Space Science.

At the Halemaʻumaʻu crater of Kīlauea, lava has sometimes filled the crater to form a lava lake. The surface of the lake cools and forms a thin solid crust. The table shows the approximate densities of some materials found in and around Hawaiʻi's volcanoes.

Material	Approximate density (g/cm³)
Pumice (a rock full of gas bubbles)	0.6
Ice	0.9
Liquid water	1.0
Molten (liquid) basalt lava	2.7
Granite (continental crust)	2.7
Solid basalt (oceanic crust)	3.0

_____ 1. Which rock would float on liquid water?
(1) solid basalt
(2) granite
(3) pumice
(4) molten basalt

_____ 2. Which statement best compares water and basalt?
(1) Both are less dense as solids than as liquids.
(2) Both are more dense as solids than as liquids.
(3) Ice is less dense than liquid water, but solid basalt is more dense than liquid basalt.
(4) Ice is more dense than liquid water, but solid basalt is less dense than liquid basalt.

3. Using the data, predict what happens to pieces of solid crust that break off the surface of the lava lake. Explain your prediction.

_____

_____

4. Use the densities in the table to explain why oceanic crust sinks beneath continental crust when the two collide.

_____

_____

Mini Glossary

Absorb (energy) – to take in energy; particles speed up

Basalt – a dark, dense igneous rock that makes up most of the oceanic crust

Condensation – the phase change from gas to liquid; energy is released

Continental crust – the thicker, less dense crust that makes up the continents, mostly granite

Contract – to take up less space; most materials contract when cooled

Crystallization – the formation of mineral crystals as magma or lava freezes

Density – the amount of mass in a given volume of a material (density = mass ÷ volume)

Evaporation – the phase change from liquid to gas; energy is absorbed

Expand – to take up more space; most materials expand when heated

Freezing – the phase change from liquid to solid; energy is released

Granite – a lighter-colored igneous rock that makes up most of the continental crust

Lava – melted rock that has reached Earth's surface

Magma – melted rock below Earth's surface

Melting – the phase change from solid to liquid; energy is absorbed

Oceanic crust – the thinner, more dense crust under the oceans, mostly basalt

Phase change – a change from one state of matter (solid, liquid, gas) to another

Release (energy) – to give off energy; particles slow down

Seismic waves – earthquake waves that scientists use to study Earth's interior